

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Zoology**Paper: ZOB-501 — Functional Anatomy of Nonchordates**

Time: Three hours

Full Marks: 70

Note: Answer six questions including Question No. 1 which is compulsory. Draw diagrams wherever necessary. The figures in the right-hand margin indicate marks.

Question 1. Answer the following parts:(a) Define the following terms: [4 × 1 = 4]

(i) Laurer's canal

(ii) Hexacanth

(iii) Medusa

(iv) Rhopalia.

(b) State whether the following statements are True or False. Give brief justification for your answer: [4 × 2 = 8](i) Polypoid phase is dominant phase in *Aurelia*.(ii) In *Fasciola* the germ cells of Cercaria gives rise to Metacercaria.(iii) Genital papilla is only found in male *Ascaris*.(iv) Coelom in *Unio* is schizocoelic.(c) Differentiate between the following: [4 × 2 = 8]

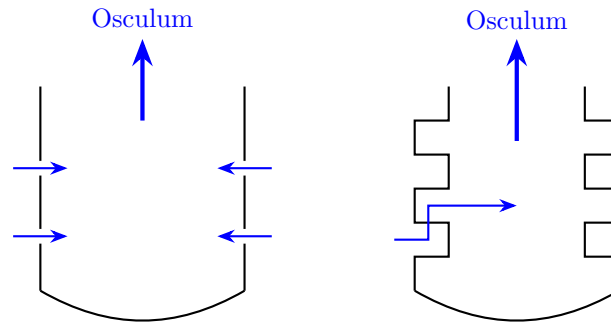
(i) Hydrozoa and Scyphozoa

(ii) Canal system and Water vascular system

(iii) Nereis and Heteronereis

(iv) Protonephridia and Nephridia.

Question 2. With the help of neat labeled diagrams, explain different types of canal system in sponges. [10]

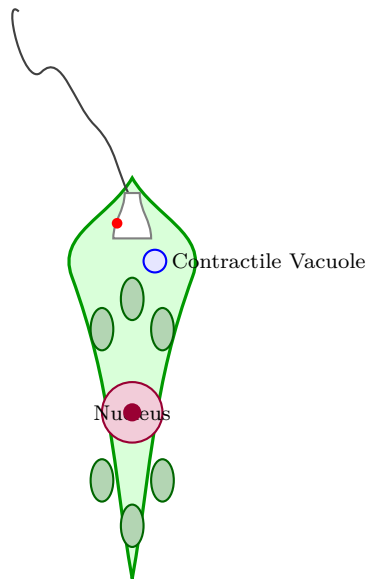


Question 3. Describe the various larval forms of *Fasciola*.

[10]

Question 4. Illustrate the life history of *Ascaris*. Add a note on its pathogenicity. [10]

Question 5. Describe the structure of *Euglena* with a neat diagram. Comment on euglenoid movement. [10]



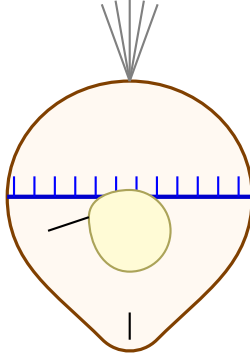
Question 6. Give a detailed account of male and female reproductive system of *Palaemon*. [10]

Question 7. Describe water vascular system of starfish and mention the role of this system in its life. [10]

Question 8. Write short notes on any two of the following:

[2 × 5 = 10]

- (a) Parasitic adaptations in *Taenia*
- (b) Trochophore larva



- (c) Torsion.

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